The origin value $D_{-\nu}(0) = \sqrt{\pi}/(2^{\nu/2}\Gamma(\frac{\nu+1}{2}))$ via Legendre duplication

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Abstract

Unit 12 of the grammar-paper §4 autoformalisation: the value of the parabolic cylinder function at the origin, $D_{-\nu}(0) = \sqrt{\pi}/(2^{\nu/2}\Gamma(\frac{\nu+1}{2}))$, as displayed in the proof of `cor:fluctuation_closed_form`. A short reconciliation of the already-proven `parCylNeg_zero` ($= 2^{\nu/2-1}\Gamma(\nu/2)/\Gamma(\nu)$, from the Γ-integral) with the paper's form, using Legendre's duplication formula. Zero `sorry`/`axiom`.

1 Setting

The `ParabolicCylinder` module (unit 6) evaluated $D_{-\nu}(0)$ directly from the DLMF integral as $2^{\nu/2-1}\Gamma(\nu/2)/\Gamma(\nu)$. The paper states the equivalent Legendre-normalised form $\sqrt{\pi}/(2^{\nu/2}\Gamma(\frac{\nu+1}{2}))$; this unit records that the two agree.

2 The thing formalised

```
theorem parCylNeg_zero_paper (  :  ) (h  : 0 <  ) :
    parCylNeg   0 = Real.sqrt   / (2 ^ (  / 2) * Real.Gamma ((  + 1) / 2))
```

3 Proof strategy

Legendre duplication at $s = \nu/2$ (`Real.Gamma_mul_Gamma_add_half`) gives $\Gamma(\nu/2)\Gamma(\frac{\nu+1}{2}) = \Gamma(\nu)\,2^{1-\nu}\sqrt{\pi}$. Cross-multiplying the goal (`div_eq_div_iff`, with $\Gamma(\nu), \Gamma(\frac{\nu+1}{2}) > 0$) reduces it to $(2^{\nu/2-1}2^{\nu/2})\big(\Gamma(\nu/2)\Gamma(\frac{\nu+1}{2})\big) = \sqrt{\pi}\,\Gamma(\nu)$; substituting the duplication identity and collapsing the powers $2^{\nu/2-1}2^{\nu/2} = 2^{\nu-1}$ and $2^{\nu-1}2^{1-\nu} = 1$ (`Real.rpow_add`) closes it by `ring`.

4 Roadblocks and resolutions

None of substance — a short algebraic reconciliation. The only care is grouping the $2^{\bullet}$ factors before `Real.rpow_add` (which does not combine powers buried in a larger product) and supplying the positivity side conditions for `div_eq_div_iff`.

5 What was Mathlib, what was new

Mathlib supplied Legendre duplication (`Real.Gamma_mul_Gamma_add_half`) and the Γ positivity/rpow lemmas. New is only the identification of the two closed forms for $D_{-\nu}(0)$.

6 Lessons

When two closed forms of a special-function value differ by a Γ-duplication, the reconciliation is a cross-multiplication plus one duplication rewrite; group the power factors explicitly so `Real.rpow_add` can fire.

7 Follow-ups

The remaining §4 items (general-k order-derivative and the log-insertion identities `prop:LogODE/lem:log_shift/lem:log_generating`; the ladder-operator commutator $[b, b^\dagger] = 1$) each need substantial new infrastructure — iterated parametric differentiation or a Weyl-algebra framework.